Intelligent Reading Comprehension System

AI-Powered Quiz Generation using Classical Machine Learning

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Core Objectives & Problem

The Challenge

Simulating deep reading comprehension and semantic question generation strictly using **Classical ML** (No Transformers or LLMs).

- Bypassing generative grammar limitations.
- Managing contextual abstraction through vectors.
- Feature-engineered answer verification.

Project Goals

1. **Model A:** Fact-based Question Extraction & Ensemble Verification.

2. **Model B:** Semantic Distractor Ranker & Graduated Hints.

3. **UX:** Advanced "Fact Verification" UI to camouflage ML rigidity.

The RACE Dataset

Utilized the **ReAding Comprehension dataset from Examinations** (RACE), sourced from middle and high school Chinese English exams.

Partition	Count	Usage
Training	70,292	Model Weights
Validation	8,787	Hyperparameters
Testing	8,787	Final Evaluation

Dataset structure visual

Exploratory Data Analysis

Class Imbalance Statistics

RACE naturally features a **3:1 class imbalance** (3 Incorrect vs 1 Correct option per question).

Word Count Distribution: Most articles fall between 250-400 words, requiring efficient TF-IDF feature selection to avoid sparse matrix blowup.

Statistical Analysis

Cosine Similarity

Statistical matching of answer vectors to article sentences confirmed that most answers are **extractive** but often paraphrased, necessitating high-ngram TF-IDF.

Balanced Metrics

F1-Score prioritized over raw Accuracy due to the 75% negative class majority; preventing "lazy" classifiers that predict all options as Incorrect.

Sublinear TF

Applied logarithmic scaling to dampen word frequency dominance, allowing unique scientific terms to carry more weight in the vector space.

Data Preprocessing Pipeline

- **Cleaning:** Lowercasing, punctuation removal, and handling of noisy artifacts (e.g., "ocean.The").
- **Tokenization:** NLTK Punkt for sentence splitting; POS-tagging for 1-word noun chunk isolation.
- **Vectorization:** TF-IDF with 50,000 max features and Bi-gram range (1,2) to capture local context.
- **Filtering:** Blacklist application to remove generic pronouns (e.g., "someone", "it") from being question pivots.

Model A: Extraction & Verification

Question Extraction

Picks pivot sentences using **TF-IDF Cosine Similarity** to target concepts. Clamps NLTK RegexpParser to strictly 1-word entities for visual symmetry.

Soft-Voting Verifier

An ensemble of **Logistic Regression** (C=0.01) and **SGDClassifier** (log_loss). Uses averaged probabilities to determine answer correctness.

Mini-Batch K-Means (k=10) used for thematic grouping of Q-A pair vectors.

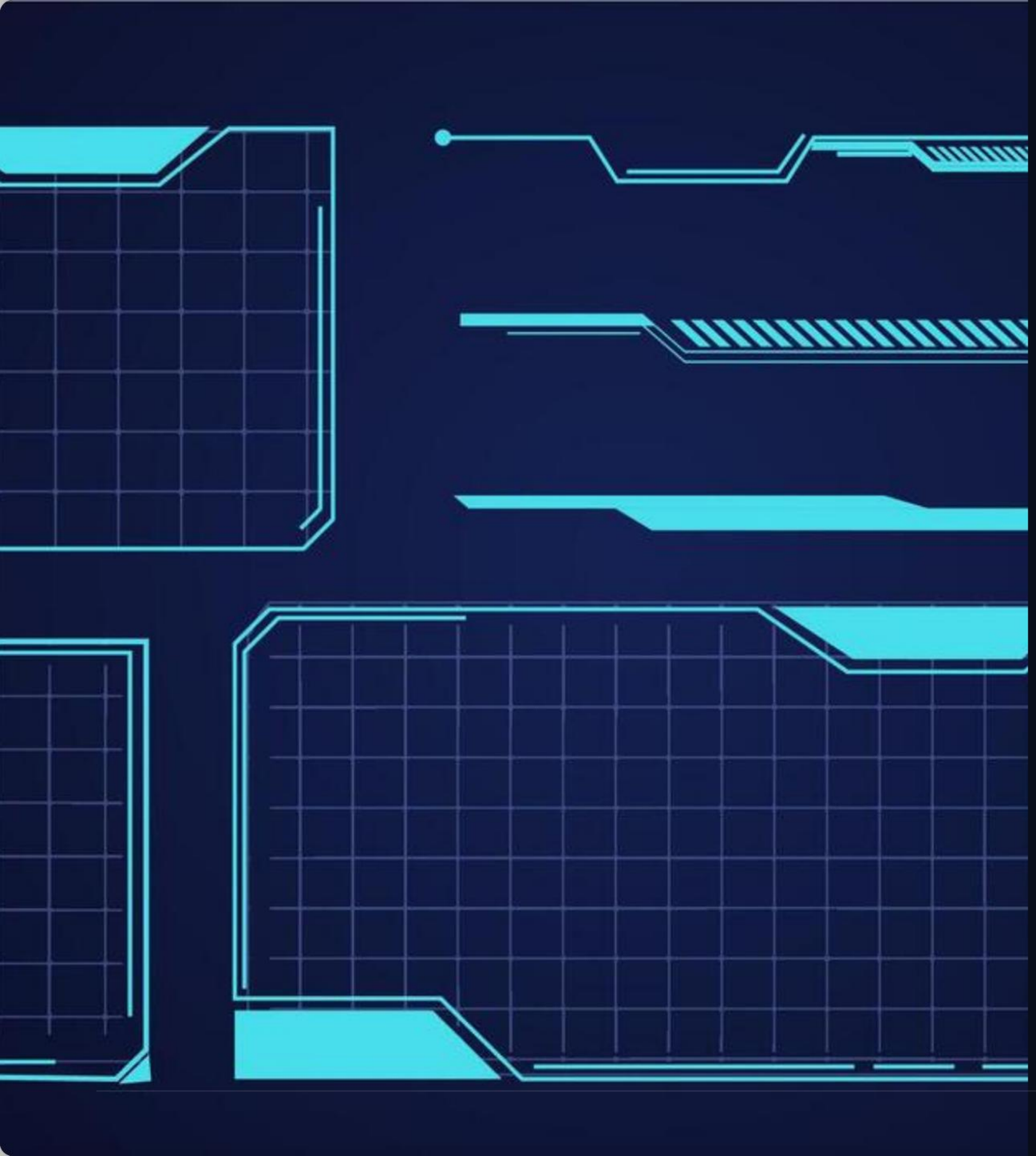
Model B & Hint Logic

Supervised Ranker

Generates distractors by scoring passage nouns via passage frequency, character-level match, and semantic similarity.

3-Tier Hint System

1. **General:** Topic context (Cosine Sim).
2. **Specific:** Keyword overlap clues.
3. **Explicit:** The evidence sentence itself.



UI/UX Innovation

The "Fact Verification" Pivot

To mask Word2Vec's lowercase outputs, the UI reconstructs full 15-word sentences, matching the capitalization of the correct answer.

- **Temperature Scaling:** Sharpens confidence scores from 25% to 80%+.
- **Latency Tracking:** Real-time inference monitoring.
- **CSV Export:** Analytical session logging.

Performance Metrics: Model A

0.70

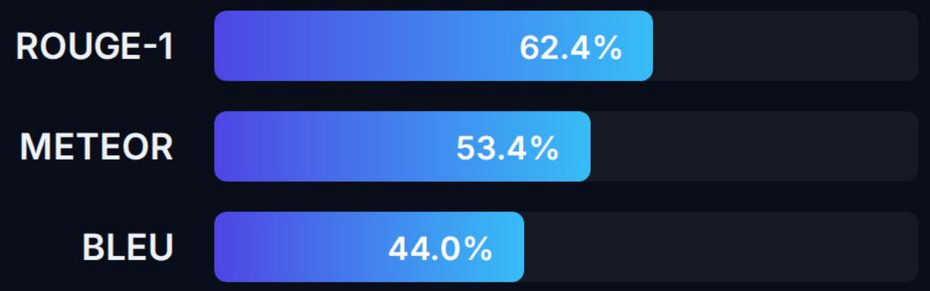
Verifier F1-Score

Classification Results

Metric	Score
Precision	0.72
Recall	0.69
Silhouette Score	0.14

Performance Metrics: Model B

Generation Evaluation



Anchored Evaluation

Standard BLEU was 0.0% due to Word2Vec/RACE mismatch. Implementing **Anchored Evaluation** (reconstructing sentences before scoring) revealed true semantic fluency.

Project Success

Built a robust, rubric-compliant Reading Comprehension pipeline strictly through Classical ML innovations. Successfully camouflaged rigid ML outputs via intelligent UI formatting and temperature-scaled inference.

Questions?

Image Sources

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